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ELE1204: Digital Electronics and Logic Design

Lecture 07 (BSCS)

Memory and Programmable Logic Devices

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Lecture Objectives and Learning outcomes

The Objectives of this lecture are :

- To understand the role of memory in computing
- To analyse the different types of memory

By the end of this lecture, students should be able to:

- Define memory and its sub categories
- Be able to differentiate between the subcategories of memory.

NOTE:: SMART = **S**pecific, **M**easurable, **A**chievable, **R**elevant, **T**imebound

Lecture Overview

- Introduction to Memory
- Types of memory
- ROM and its types
- **RAM and its types**
- SRAM and DRAM

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Introduction to Memory

Memory is a crucial component in computer systems and digital devices, used to **store, retrieve, and manage data**. It plays a vital role in processing operations, enabling fast access to instructions and information. Memory can be broadly classified into **volatile** and **non-volatile** memory.

Importance of Studying Memory in Digital Systems

- Understanding memory organization improves system performance.
- Different memory types have specific roles in computing and embedded systems.
- Memory selection impacts cost, speed, and power efficiency in device design.

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Types of Memory

Memory is categorized into two main types:

1 Volatile Memory (RAM - Random Access Memory)

- Data is lost when power is turned off.
- Used for **temporary storage** during processing.
- Faster access compared to non-volatile memory.

2 Non-Volatile Memory (ROM - Read-Only Memory)

- Data is **permanently stored**, even without power.
- Used for **firmware and system boot processes**.
- Cannot be modified easily (except for certain types like EEPROM).

ROM (Read-Only Memory) and Its Types

- **ROM (Read-Only Memory)**

Stores **permanent data** that is programmed during manufacturing.
Cannot be modified after production.
Example: BIOS (Basic Input/Output System) in computers.

- **PROM (Programmable Read-Only Memory)**

Initially **blank**, but can be programmed **once** using a PROM burner.
Cannot be erased or modified after programming.
Example: Used in **embedded systems** and **game cartridges**.

- **EPROM (Erasable Programmable Read-Only Memory)**

Can be erased using **ultraviolet (UV) light** and reprogrammed.

Used in older computer systems and embedded applications.
Example: Used in **legacy BIOS chips** before EEPROM became common.

EEPROM (Electrically Erasable Programmable Read-Only Memory)

Can be erased and reprogrammed **electrically**, one byte at a time.

Used in **modern BIOS, flash drives, and memory cards**.
Example: Used in **firmware updates and flash storage (SSD, USB drives)**

3. RAM (Random Access Memory) and Its Types

RAM is volatile memory used for **temporary storage** during CPU operations.

Types of RAM

Type	Full Form	Speed	Power Consumption	Used In
SRAM	Static RAM	Faster	Low	CPU Cache, Registers
DRAM	Dynamic RAM	Slower	High	Main Memory (RAM Modules)

Difference Between SRAM and DRAM

Feature	SRAM	DRAM
Speed	Faster	Slower
Power Consumption	Low	High (Needs constant refreshing)
Complexity	More complex (Uses flip-flops)	Simpler(Uses capacitors)
Cost	Expensive	Cheaper
Usage	CPU cache, registers	Main memory(RAM modules)

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Thank You

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